

Dataset Guide

Volatility and tail-risk forecasting across six Tier-1 equity indices

GARCH-EVT · Realized GARCH · Quantile Regression

Item	Value
Dataset	GARCH-EVT vs Realized GARCH vs Quantile Regression - Researcher A
Generated	2026-08-23
Primary file	01_ANALYSIS_READY/<CODE>_analysis.csv (96 columns)
Primary sample	Sample B: all six indices, 2013-09-30 to 2026-08-21, 2,685 common days
Indices	SPX, NDX, UKX, DAX, NKY, HSI
Daily coverage	1990-01-02 to 2026-08-21
Realized coverage	2011-09 onward (DAX 2013-09); NKY has no valid RV in 2016-17
Validation	1,195 acquisition checks + 158 analysis checks, 0 failures
Models fitted	NONE. This is a dataset only - no model has been estimated.

1. Quick start

Model on 01_ANALYSIS_READY/<CODE>_analysis.csv. That is the only folder the modelling code should read. Everything else is raw input, intermediate output, or documentation.

A minimal, correct load looks like this — note the two things it does that are easy to get wrong: it gates the realized measures on RV_Valid, and it applies the forecasting lag itself.

```
import pandas as pd
a = pd.read_csv('01_ANALYSIS_READY/SPX_analysis.csv', parse_dates=['Date'])

# daily-only models (GARCH-EVT, Quantile Regression) can use all history
r = a.loc[a['Return'].notna(), ['Date', 'Return']]

# realized measures MUST be gated
rv = a.loc[a['RV_Valid'] & a['InSample_B'], ['Date', 'RV', 'LogRV', 'RS_neg']]

# YOU apply the lag - nothing in the file is pre-lagged
X = a[['LogRV', 'LogRV_w', 'LogIV', 'NegReturn']].shift(1)
y = a['Return']
```

2. Folder map

Folder	Contents
00_DOCUMENTATION	Reports, PDFs, dictionaries. START HERE.
01_ANALYSIS_READY	**THE dataset to model on.** One cleaned CSV per index, 96 columns.
02_RAW_DAILY	Exchange daily OHLC per index, 1990-2026. Yahoo Finance.
03_RAW_INTRADAY	1-min and 5-min session bars per index-year. Dukascopy.
04_RAW_VOLATILITY	17 implied-volatility indices. CBOE, STOXX, Yahoo.
05_RAW_MACRO	22 keyless macro / risk-factor series. Yahoo Finance.
06_REALIZED_MEASURES	Daily realized measures per index, base and extended.
07_PANEL_INTERMEDIATE	The raw join, BEFORE cleaning. Provenance only - do not model on this.
08_VALIDATION	Every validation and EDA table behind the reports.
09_FIGURES	Ten diagnostic figures.
10_SCRIPTS	Every script, numbered in execution order.
11_LOGS	Manifests and per-phase run summaries.
12_CACHE_REGENERATION	Raw Dukascopy .npy cache. Only needed to rebuild realized measures.

3. Precautions — read before modelling

Twenty-four traps, ordered by severity. The first four will invalidate results outright if ignored; the rest will bias or misrepresent them.

CRITICAL (4)

1	Model on 01_ANALYSIS_READY, never 07_PANEL_INTERMEDIATE The panel folder is the raw join kept for provenance. It has no session-quality gating, so it still contains the Nikkei's biased 2016-17 realized variances.
2	Gate every realized measure on RV_Valid RV, RS_pos/neg, BPV, MedRV, RQ, TQ, Jump, ContVar and all HAR terms are NaN outside RV_Valid. If you fillna() or dropna() carelessly you will either reintroduce bad data or silently delete good return observations.
3	Apply the forecasting lag yourself - the file is contemporaneous Every predictor is dated at the close of day t. Nothing is pre-lagged. The modelling code must shift predictors to forecast t+1. Failing to do this produces spectacular in-sample results and is the single most common way to invalidate a volatility paper.
4	Do not winsorize, trim or de-jump the returns The tail is the object of study. Clipping extremes shrinks the estimated GPD shape parameter and manufactures VaR that appears well-calibrated only because the exceedances were deleted. The extreme days were individually verified as real market events.

HIGH (7)

5	Never use LogRV and LogRS_neg together $RS_pos + RS_neg = RV$ identically and the downside share sits at 0.50, so LogRS_neg is LogRV plus an almost-constant. Mean VIF is about 95. Use the level-plus-share form: LogRV with RSV_Ratio, JumpShare and RSkew. Max VIF falls from 21.9 to 8.1.
6	Do not use US10Y_pct or TermSpread_pct in levels as regressors Both fail ADF on all six indices ($p = 0.87$ and 0.33). Use US10Y_diff and TermSpread_diff, which are stationary at $p < 0.001$. The levels are retained only as regime descriptors.
7	BalancedRV_B is for POOLED statistics only It marks the 1,994 dates where all six indices have valid RV. Using it for per-index work discards 5,909 perfectly good index-days to accommodate the worst index. Per-index estimation and per-index DM/MCS should use each index's own RV_Valid days.
8	The Nikkei has no valid realized measure in 2016-17 The Dukascopy feed is missing the entire 09:00 opening hour for those two years. A rolling window must use the last AVAILABLE observations rather than the last N calendar days, and no Realized-GARCH forecast can be produced or evaluated inside the gap. Disclose it.
9	RV measures the cash session only - it is not daily variance The daily return spans close to close and also contains the overnight gap. RV captures only 33-58% of daily variance depending on the market. Use ScaleFactor_HL (1.71 to 3.04, index-specific) or RV_Scaled when comparing RV against squared daily returns, and let the Realized-GARCH measurement equation absorb the scale.
10	ScaleFactor_HL is a full-sample, in-sample constant It is estimated once over the whole of sample B. Fine for description and for the measurement equation; for a strict recursive out-of-sample exercise re-estimate it on each rolling window or you have leaked information.
11	Sessions are not synchronous - do not pool naively on the calendar date Tokyo and Hong Kong close before New York opens, so a US shock on day t reaches Asia on day t+1. Same-date return correlation between Asia and the US is only about 0.18, which is a timing artefact and not economic independence. Either adopt a lagged-information convention or estimate index by index and pool only the loss series.

MEDIUM (8)

12	Re-estimate the POT threshold on the actual GARCH residuals The recommended 95th-97.5th percentile comes from rolling-standardised returns, used as a stand-in because no GARCH was fitted at the EDA stage. McNeil-Frey applies the GPD to genuine GARCH residuals; redo the stability plot once stage 1 exists.
13	Volatility indices are annualised percentages, RV is a daily variance Convert before comparing: $IV_DailyVar = (VolIdx/100)^2 / 252$. This is already provided. Mixing the two scales silently is an easy and expensive mistake.
14	Overnight_LogRet is the only column not dated at the close It spans close(t-1) to open(t) and is therefore known at the OPEN of day t. Treat it as the first observable piece of day t, not as day-t close information.
15	Macro series are forward-filled up to 5 business days These are US series joined onto six exchange calendars. The fill is forward only, never backward, so it is the information set rather than an interpolation - but check MacroFilled_t if a result depends on a specific date.

16	Half-days are real sessions but are excluded from RV_Valid A three-hour variance fed into a measurement equation calibrated on full sessions biases the intercept. They are retained in the file with IsHalfDay=True, so the choice is reversible if you want them.
17	The intraday data is a bid-side index CFD, not the exchange index Aligned correlation against the exchange daily return is 0.97-0.99, so it tracks well, but it is a proxy and must be described as one. The 1-min/5-min RV ratio of 1.06-1.09 shows a smoothed feed, which also means RV may be mildly damped versus a trade-based estimator.
18	Three volatility indices are regional proxies, not the index's own NKY uses VXEFA in the primary sample (the Nikkei VI only starts 2018), UKX uses VXEFA and HSI uses VXEEM, because no free FTSE-100 or HSI volatility index exists. Volldx_IsProxy flags the Nikkei case. Sample C exists to test that this does not drive results.
19	Sample B excludes 2008 The daily-only models estimate from 1990 and see the GFC; Realized GARCH cannot, because free intraday history begins in 2011-2013. The three models therefore see different amounts of crisis history. Unavoidable, but it must be stated.

LOW (5)

20	Use Close only from the macro folder on OHLC-inconsistent rows Yahoo writes placeholder bars for FX and front-month futures on a minority of days where Open=High=Low and Close comes from a different snapshot. Close is the reliable field. OHLC_Consistent flags the affected rows.
21	Do not recompute returns from 07_PANEL_INTERMEDIATE The download rounded LogReturn to 6 decimal places. The analysis files re-derive the return from Close at full precision; use those.
22	If you re-download from Dukascopy: months are ZERO-INDEXED in the URL January is 00. Getting this wrong silently returns the wrong month rather than an error.
23	If you re-download: one process only, with HTTP keep-alive Dukascopy throttles per IP. Extra parallel processes made throughput worse (4.9 to 1.9 req/s) and triggered a lasting penalty. Without keep-alive it is ~0.05 req/s, a 100x penalty.
24	An expired CA bundle silently breaks every HTTPS fetch This cost real time at the start. If downloads fail wholesale, update certifi before debugging anything else.

4. Sample definition

Sample	N_Indices	Indices	Start	End	Binding_Index	Common_Days	Approx_Years	Note
A	5	SPX NDX UKX NKY HSI	2011-09-20	2026-08-21	NKY	3056	12.13	5 indices, no euro area, longest window. Robustness.
B	6	SPX NDX UKX DAX NKY HSI	2013-09-30	2026-08-21	DAX	2685	10.65	PRIMARY. All six. NKY on the declared VXEFA proxy.
C	6	SPX NDX UKX DAX NKY HSI	2018-01-22	2026-08-21	NKY	1892	7.51	All six on their own vol index, from the Nikkei VI start. Robustness.

Sample B is primary. It keeps all six indices and therefore the euro area, which sample A drops. Dropping the DAX would also remove the only non-US market with a native model-free volatility index (VDAX-NEW), leaving every European and Asian result resting on a proxy. Sample C spends a third of the observations to remove one declared proxy, which is a bad trade. Both A and C are retained as ready-made robustness runs.

Three flags encode the distinction and they are not interchangeable:

Flag	Meaning
InSample_B	The window. Use for per-index work.
CommonDate_B	2,685 dates on which all six indices traded.
BalancedRV_B	1,994 dates where all six ALSO have valid RV. Pooled statistics only.

5. Cleaning decisions

#	Decision	Reason	Effect
1	Realized measures nulled on DEFECT sessions; rows retained	The exchange close is still valid, so the daily return is still an observation. Deleting rows would shorten the return series non-randomly with respect to volatility.	RV NaN on 369/155/80/24/633/71 days (SPX/NDX/UKX/DAX/NKY/HSI)
2	Half-days flagged and excluded from RV_Valid; values retained	A short-session variance biases a measurement equation calibrated on full sessions. Standard practice in the realized-volatility literature.	Reversible via IsHalfDay
3	Returns NOT winsorized, trimmed or de-jumped	The tail is the object of study. Clipping would shrink the GPD shape parameter and delete the exceedances that identify it.	Extremes individually verified as real events
4	Macro forward-filled <=5 business days, forward only	US series on six exchange calendars. The last published value IS the information set on a day the US did not trade. Backward fill would be look-ahead.	MacroFilled_t records it
5	Hansen-Lunde scale factor estimated per index	The session vs close-to-close gap is index-specific and large. Measured, not assumed away.	1.71 DAX to 3.04 NKY, monotone in session length
6	Nothing standardised, differenced or de-meanned in storage	Those are estimator-specific choices. Levels are stored; transformations belong in the modelling code.	-
7	Non-stationary macro levels supplemented with differences	ADF $p=0.87$ for the 10-year yield level.	US10Y_diff, TermSpread_diff added
8	Realized block re-parameterised as level + share	LogRV and LogRS_neg are collinear by identity.	Max VIF 21.9 -> 8.1
9	Session coverage measured on the 5-minute grid, not the 1-minute grid	A quiet minute with no quote change is dropped upstream and is not missing data. The 1-minute version called 674 SPX days half-days against ~3/year scheduled.	Half-days now land on 07-03, 07-04, 12-24, 12-31
10	Return re-derived from Close at full precision	The download rounded LogReturn to 6 dp.	Removes a 5e-7 quantisation

6. Data quality findings

6.1 Structural integrity

No duplicated dates, no weekend dates, no OHLC ordering violations, no zero realized variances and no runs of repeated closing prices, on any index. Zero-return frequency peaks at 0.15%.

6.2 The Nikkei intraday defect — the finding that drove the cleaning

The Dukascopy feed carries **no bars at all in the 09:00 local hour for the whole of 2016 and 2017**. Because intraday volatility is U-shaped, losing the opening hour removes far more than its proportional share of the session's variance, and the correction factor could only be estimated from the very period that is missing.

The bias was measured, not assumed, against two benchmarks that do not depend on intraday coverage — the squared daily return and the Parkinson high-low estimator from the exchange daily bar:

Index	Session class	n	RV / Parkinson (median)
NKY	FULL	2,265	1.030
NKY	DEFECT	598	0.786
SPX (control)	FULL	3,113	1.135
SPX (control)	DEFECT	91	1.035

Realized variance on Nikkei defect days runs about **24% below** where it should. Those days are nulled; the rows are kept, because the exchange close is still valid and the daily return is still a good observation.

6.3 Session classification

Symbol	DEFECT	FULL	HALFDAY	Total	Pct_DEFECT
DAX	30	3280	2	3312	0.91
HSI	57	3497	32	3586	1.59
NDX	122	3496	119	3737	3.26
NKY	672	2869	8	3549	18.93
SPX	346	3297	117	3760	9.2
UKX	81	3641	17	3739	2.17

Half-days were separated from feed defects by *where* the missing 5-minute blocks sit — a half-day is a contiguous truncation at the end, a defect is a missing open or an interior hole. The confirmation that this works is that the half-days land on exactly the dates the exchanges schedule: 07-03 and 07-04 for the US indices, 12-24 and 12-31 for London and Hong Kong.

7. Statistical properties and what they imply

Every test below exists to justify — or refuse to justify — a specific modelling choice.

7.1 Returns are far from Gaussian

Code	N	Mean	SD	Skew	ExcessKurt	Min	P1	P99	Max
SPX	9226	0.0003317	0.01134	-0.364	10.87	-0.1277	-0.03154	0.02982	0.1096
NDX	9226	0.0005265	0.0166	-0.01216	6.315	-0.13	-0.04616	0.0443	0.172
UKX	9254	0.0001612	0.01073	-0.3068	7.801	-0.1151	-0.03102	0.0281	0.09384
DAX	9269	0.0002893	0.01369	-0.2032	5.873	-0.1305	-0.03999	0.03456	0.108
NKY	8989	5.938e-05	0.01491	-0.1704	5.926	-0.1323	-0.04033	0.03763	0.1323
HSI	9040	0.0002451	0.01554	-0.07965	9.184	-0.1473	-0.04247	0.04054	0.1725

7.2 Tail index — the case for EVT

Code	k	Hill_Alpha	SE
SPX	213	2.982	0.2043
NDX	208	3.618	0.2509
UKX	220	3.242	0.2186
DAX	217	3.147	0.2136
NKY	219	3.384	0.2286
HSI	218	2.966	0.2009

Hill alpha runs 2.6 to 3.9. **Every estimate is below 4**, so the fourth moment does not exist and the sample kurtosis is not estimating any finite population quantity. Several are close to 3, putting the third moment in doubt. A Gaussian-innovation GARCH assumes all moments exist; a Student-t GARCH imposes one tail parameter on both tails and on the whole distribution at once. Neither is consistent with alpha near 3. **This is the quantitative case for the EVT stage.**

7.3 Specification tests

Code	ADF_Return_p	KPSS_Return_p	LB10_Return_p	LB22_RetSq_p	ARCH_LM10_p	EngleNg_p
SPX	4.014e-30	0.1	2.768e-15	0	0	1.091e-13
NDX	1.821e-27	0.1	3.317e-12	0	0	1.517e-17
UKX	7.408e-30	0.1	4.857e-11	0	0	3.994e-24
DAX	0	0.1	0.1007	0	0	6.649e-05
NKY	0	0.0102	0.01716	0	6.483e-288	4.366e-28
HSI	0	0.1	0.00822	0	0	4.625e-06

Result	What it decides
ADF rejects, KPSS does not	Returns are stationary; model in levels.
Ljung-Box on r rejects (5 of 6)	A conditional-mean term is warranted; AR(1) suffices.
ARCH-LM rejects at $p < 1e-16$	A conditional-variance model is mandatory.
Engle-Ng rejects on all six	Plain GARCH(1,1) is misspecified. Use GJR or EGARCH.

7.4 Realized-measure dynamics

Cod e	ADF_LogRV_p	KPSS_LogRV_p	GPH_d_LogRV	AC1_LogRV	AC22_LogRV	AC66_LogRV	Leverage_corr_r_LogRV_next	VarShare_Session_Pct
SPX	4.316e-09	0.01	0.5706	0.7867	0.3617	0.1454	-0.1924	55.29
NDX	3.989e-08	0.01	0.5702	0.7748	0.3877	0.1643	-0.203	55.48
UKX	9.169e-11	0.05164	0.6262	0.77	0.3816	0.2482	-0.1758	57.53
DAX	6.683e-14	0.05074	0.5012	0.7412	0.3545	0.2395	-0.1848	58.42
NKY	1.225e-07	0.01456	0.6189	0.6731	0.2661	0.06581	-0.1579	32.9
HSI	9.226e-08	0.01	0.5074	0.7061	0.3976	0.2862	-0.1198	46.71

The GPH fractional-integration estimate is 0.50–0.63 and ADF and KPSS *both* reject — the classic long-memory signature rather than a clean $I(0)$ or $I(1)$. This is the case for the HAR cascade. The leverage correlation between today's signed return and tomorrow's log realized variance is negative for every index, the same asymmetry Engle-Ng detects.

7.5 The scale gap

Code	ScaleFactor_HL	Median_RVol_Ann	Median_VolIdx
SPX	1.808	8.49	16.24
NDX	1.803	11.75	19.83
UKX	1.738	8.95	16.29
DAX	1.712	10.95	18.33
NKY	3.039	9.79	16.23
HSI	2.141	12.26	20.64

The scale factor rises monotonically as the cash session shortens — 8.5-hour markets at 1.71 and 1.74, the 6.5-hour US indices at 1.80, Hong Kong at 2.14 and Tokyo at 3.04. That ordering is the internal check that the number is real rather than an artefact. Tokyo's overnight window contains the entire US session, which is why two thirds of its daily variance is invisible to the intraday data.

8. Predictor screening

Averaged over the six indices. All predictors dated t , all targets $t+1$ — these are forecasting numbers, not contemporaneous correlations.

Best for next-day log realized variance

Predictor	R2_LogRV_next	PseudoR2_q05	PseudoR2_q01
LogRV_w	0.5671	0.0943	0.1564
LogRV	0.5531	0.1045	0.1813
LogRS_neg	0.5158	0.09947	0.1786
LogRV_m	0.4251	0.06375	0.09305
LogIV	0.4219	0.1049	0.1814
RangePct	0.4169	0.08405	0.1657
VolIdx	0.4093	0.1107	0.1983
IV_DailyVar	0.3244	0.1055	0.1986
RSneg_w	0.2914	0.08749	0.179
RV_w	0.2885	0.08713	0.1788

Best for the 1% left tail of next-day returns

Predictor	R2_LogRV_next	PseudoR2_q05	PseudoR2_q01
IV_DailyVar	0.3244	0.1055	0.1986
ContVar	0.2495	0.08976	0.1983
VolIdx	0.4093	0.1107	0.1983
RV	0.2497	0.08756	0.1928
LogIV	0.4219	0.1049	0.1814
LogRV	0.5531	0.1045	0.1813
RSneg_w	0.2914	0.08749	0.179
RV_w	0.2885	0.08713	0.1788
LogRS_neg	0.5158	0.09947	0.1786
RangePct	0.4169	0.08405	0.1657

The ranking flips. For forecasting the level of volatility the realized measures win; for the 1% tail the implied-volatility index wins. The option market carries forward-looking information about extreme outcomes that the backward-looking realized measures do not. That is a genuine result and a natural thing for the paper to report.

RangePct, computed from the exchange daily high-low alone, reaches $R^2 = 0.42$ — close to the implied-vol index — and is available on every day including those where the intraday feed failed. It is the natural fallback regressor across the Nikkei's 2016-17 gap.

Collinearity

Parameterisation	Max mean VIF	Verdict
LogRV + LogRS_neg + HAR terms	21.9	Unusable — RS_pos + RS_neg = RV identically
LogRV + RSV_Ratio + JumpShare + RS skew + HAR	8.1	Use this parameterisation

9. Feature sets by model

Model	Role	Fields	Note
GARCH-EVT	target	Return	log return, full history from 1990 - this model needs no intraday data
GARCH-EVT	estimation note	-	stage 1 AR(1)-GJR-GARCH on Return; stage 2 GPD on the standardised residuals

Model	Role	Fields	Note
GARCH-EVT	POT threshold	-	95th-97.5th percentile of the standardised residuals - the region where the stage-6 stability plot shows xi flat with 230-460 exceedances
GARCH-EVT	sample	InSample_B for evaluation	estimate on all history; only the forecast window must match the other models
Realized GARCH	target	Return	log return
Realized GARCH	realized measure	RV	5-min RV, gated on RV_Valid. RV_ss is the low-noise alternative
Realized GARCH	measurement eq.	ScaleFactor_HL	RV covers the session only, so it is 1.71-3.04x below close-to-close variance depending on session length; the intercept must absorb this or RV_Scaled must be used
Realized GARCH	optional	MedRV, ContVar, Jump, RS_neg	jump-robust and signed variants for the robustness table
Realized GARCH	sample	RV_Valid & InSample_B	NKY has no valid RV in 2016-17; that index is estimated on 2,265 days with a gap
Quantile Regression	target	Return(t+1)	next-day log return, tau = 0.01 and 0.05
Quantile Regression	core regressors	LogRV, LogRV_w, LogRV_m, LogIV, VRP, NegReturn, RangePct	the orthogonal parameterisation - max VIF 8.1 versus 21.9 for the naive set
Quantile Regression	asymmetry	RSV_Ratio, JumpShare, RSkew	downside share, jump share and intraday skew; these replace LogRS_neg, which is collinear with LogRV by construction
Quantile Regression	05_RAW_MACRO	TermSpread_diff, CreditStress, DXY_ret	differenced or already-stationary forms only
Quantile Regression	fallback	RangePct, ParkinsonVar, GarmanKlassVar	available on every day including intraday-defect days; RangePct alone reaches R2=0.42 for next-day log RV, close to the implied-vol index
Quantile Regression	excluded	LogRS_neg, US10Y_pct, TermSpread_pct level	collinear (VIF 95) or non-stationary in levels
All models	evaluation	RV or RV_Scaled as the volatility proxy	QLIKE and MSE against the realized measure; VaR backtests against Return
All models	pooled tests	BalancedRV_B	Model Confidence Set across indices requires aligned loss series

10. EVT threshold selection

GPD shape parameter ξ fitted to the left tail of **rolling-standardised** returns. Standardising first is what makes this the right diagnostic: McNeil-Frey applies the GPD to GARCH residuals, not raw returns, because the limit theory assumes independence.

q	DAX	HSI	NDX	NKY	SPX	UKX
0.8	0.03415	0.09908	0.03647	0.07359	0.08612	0.08238
0.85	0.05625	0.1444	0.05894	0.08394	0.1065	0.1037
0.9	0.07067	0.2144	0.09342	0.1331	0.1402	0.1157
0.925	0.0992	0.2258	0.08779	0.1319	0.1704	0.1141
0.95	0.1632	0.2071	0.1185	0.1534	0.1753	0.1429
0.96	0.1681	0.1808	0.09846	0.2019	0.1436	0.1349
0.97	0.187	0.1964	0.1369	0.196	0.1739	0.1545
0.975	0.2221	0.1957	0.1586	0.2768	0.181	0.1257
0.98	0.2451	0.2268	0.1735	0.2521	0.2296	0.171
0.99	0.2095	0.1716	0.2285	0.1833	0.3805	0.1899

Exceedance counts at the same thresholds

q	DAX	HSI	NDX	NKY	SPX	UKX
0.8	1,834	1,788	1,825	1,778	1,825	1,831
0.85	1,376	1,341	1,369	1,334	1,369	1,373
0.9	917	894	913	889	913	916
0.925	688	671	685	667	685	687
0.95	459	447	457	445	457	458
0.96	367	358	365	356	365	367
0.97	276	269	274	267	274	275
0.975	230	224	229	223	229	229
0.98	184	179	183	178	183	184
0.99	92	90	92	89	92	92

ξ is positive throughout — a heavy, Fréchet-domain tail that *survives* volatility standardisation, which is precisely the McNeil-Frey argument for a second EVT stage on top of the GARCH filter. ξ drifts upward at low thresholds where the GPD approximation has not bitten, and becomes unstable above the 98th percentile where fewer than 190 exceedances remain. **Recommended POT threshold: the 95th to 97.5th percentile**, giving roughly 230-460 exceedances per index. Re-estimate on the actual GARCH residuals once stage 1 is fitted.

11. Known limitations

Limitation	Detail
Nikkei RV gap	No valid realized measure in 2016-17. 2,265 valid days with a hole.
Intraday is a CFD proxy	Bid-side index CFD, not the exchange index. Aligned correlation 0.97-0.99.
Proxy volatility indices	NKY uses VXEFA in sample B; UKX uses VXEFA; HSI uses VXEEM. No free native index exists for FTSE or HSI.
Sample B excludes 2008	The three models see different amounts of crisis history.
Non-synchronous sessions	Asia closes before the US opens; same-date correlation understates the true linkage.
Five FRED series absent	CPIAUCSL, UNRATE, INDPRO, NFCI, USREC have no keyless market analogue. Optional covariates; need a free key.

12. Verification

The analysis dataset passes **158 independent checks with 0 failures**, re-derived from scratch rather than re-read from the objects that produced them. A validation that reuses the build code cannot catch a bug in the build code.

The look-ahead test deserves a note. An earlier version compared each predictor's correlation with today's realized variance against its correlation with tomorrow's, and flagged anything higher on the future. That test was invalid — VRP, JumpShare and RSV_Ratio all contain RV(t) by construction, which mechanically suppresses their contemporaneous correlation, and the macro series legitimately lead the Asian indices by a day. It was replaced with a **prefix-stability test**: every time-dependent column is rebuilt from data truncated at t and compared at the cut. If a column used information from after t, truncation must change it. All six indices pass at four cut points.

Acquisition-stage validation was 1,195 checks across 239 files, also with zero failures. Mechanical identities (RS_pos + RS_neg = RV, ContVar + Jump = RV) hold to floating-point precision.

13. Reproduction

```
pip install -r requirements.txt

# acquisition
python 10_SCRIPTS/01_download_daily.py
python 10_SCRIPTS/02_download_volatility.py
python 10_SCRIPTS/03_download_intraday.py SPX NDX UKX DAX NKY HSI
python 10_SCRIPTS/09_download_macro_yahoo.py
python 10_SCRIPTS/05_build_intraday_and_RV.py
python 10_SCRIPTS/10_build_master_panel.py

# cleaning and EDA
python 10_SCRIPTS/11_define_samples.py
python 10_SCRIPTS/12_extended_realized_measures.py
python 10_SCRIPTS/13_eda_quality_audit.py
python 10_SCRIPTS/14_session_classification.py
python 10_SCRIPTS/15_build_analysis_dataset.py
python 10_SCRIPTS/16_eda_stylized_facts.py
python 10_SCRIPTS/17_eda_predictor_screening.py
python 10_SCRIPTS/18_eda_tails_breaks_features.py
python 10_SCRIPTS/19_eda_figures.py
python 10_SCRIPTS/20_finalise_and_document.py
python 10_SCRIPTS/21_build_eda_report.py
python 10_SCRIPTS/22_validate_analysis.py
python 10_SCRIPTS/23_build_documentation.py
```

Run from inside the Datasets folder. Steps 12 and 14 read the 24,000-file Dukascopy cache and take several minutes each; step 3 is resumable and only fetches what is missing. Everything else runs in seconds.

14. Data dictionary

The analysis files carry 96 columns. Grouped by role:

Columns	Group	Units	Availability	Timing	Note
Date	key	date	always	-	exchange trading date, local
Symbol	key	code	always	-	index code
Open,High,Low,Close	price	index points	always	known at close t	exchange daily bar from Yahoo Finance
Return	target	log return	always	known at close t	$\log(C_t/C_{t-1})$. PRIMARY TARGET for GARCH-EVT and Quantile Regression
Return_Pct,Return_Sq,AbsReturn, NegReturn	target	various	always	known at close t	transformations of Return; NegReturn = $ \min(r,0) $ leverage term
RV	realized	daily variance	RV_Valid only	known at close t	5-min realized variance, cash session only. PRIMARY realized measure
RVol,LogRV,RVol_Ann_Pct	realized	sd / log / annualised %	RV_Valid only	known at close t	transformations of RV
RV_1min,RV_ss	realized	daily variance	RV_Valid only	known at close t	1-min RV and the 5-grid subsampled RV (Zhang-Mykland-Ait-Sahalia)
BPV,MedRV	realized	daily variance	RV_Valid only	known at close t	jump-robust integrated variance. MedRV is robust to consecutive jumps
Jump,ContVar,JumpShare	realized	daily variance / ratio	RV_Valid only	known at close t	$J = \max(RV - MedRV, 0)$; $ContVar = RV - J$; $JumpShare = J/RV$
RS_pos,RS_neg,SignedJump,RSV_Ratio	realized	daily variance / ratio	RV_Valid only	known at close t	realized semivariance. $RS_pos + RS_neg = RV$ EXACTLY. $RSV_Ratio = RS_neg/RV$
LogRS_neg	realized	log variance	RV_Valid only	known at close t	DO NOT use together with LogRV - VIF ~95, they are the same variable plus a constant
RQ,TQ,RV_RelSE	realized	quarticity / ratio	RV_Valid only	known at close t	realized quarticity for HAR-Q; $RV_RelSE = \sqrt{2 \cdot RQ/n} / RV$ is the relative measurement error
RSkew,RKurt	realized	dimensionless	RV_Valid only	known at close t	intraday realized skewness and kurtosis (Amaya et al. 2015)
NoiseRatio_1v5	diagnostic	ratio	RV_Valid only	known at close t	RV_1min / RV_5min ; a microstructure-noise indicator, ~1.06-1.09 here
RV_d,RV_w,RV_m	HAR	daily variance	RV_Valid only	known at close t	HAR cascade: today, trailing 5-day mean, trailing 22-day mean, all INCLUDING t
LogRV_d,LogRV_w,LogRV_m,RSneg_*,CV_*,J_*	HAR	various	RV_Valid only	known at close t	same cascade on the log, semivariance, continuous and jump parts
RV_Scaled,ScaleFactor_HL	realized	daily variance	RV_Valid only	IN-SAMPLE constant	Hansen-Lunde scaling of session RV onto close-to-close variance. WARNING: the factor is estimated once over the whole of sample B, so it is in-sample information. Fine for the measurement equation and for description; for a strict recursive out-of-sample exercise re-estimate it on each rolling window.
Volldx,Volldx_Symbol,Volldx_IsProxy	implied	annualised vol %	always	known at close t	regional implied volatility index actually used; IsProxy flags NKY->VXEFA
IV_DailyVar,LogIV	implied	daily variance	always	known at close t	$(Volldx/100)^{2/252}$ - the vol index converted onto the RV scale
VRP	implied	daily variance	RV_Valid only	known at close t	variance risk premium = $IV_DailyVar - RV$
ParkinsonVar,GarmanKlassVar,RangePct	range	daily variance / %	always	known at close t	range estimators from the exchange daily bar. AVAILABLE ON EVERY DAY, including intraday-defect days - the natural fallback predictor
Overnight_LogRet	return	log return	RV_Valid only	known at OPEN of t	$close(t-1) \rightarrow open(t)$. NOT a day-t close variable; the only column that is not close-dated
CFD_SessionReturn	diagnostic	log return	RV_Valid only	known at close t	close-to-close return of the CFD used to build RV; for reconciliation against Return
US10Y_pct,US13W_pct,TermSpread_pct	05_RAW_M ACRO	percent	always	known at close t	LEVELS ARE NON-STATIONARY (ADF p=0.87/0.33). Do not use as QR regressors - use the _diff forms
US10Y_diff,TermSpread_diff	05_RAW_M ACRO	percentage points	always	known at close t	first differences; stationary at p<0.001. USE THESE in the quantile regression
DXY,WTI_usd,GOLD_usd,HYG_px,IEF_px	05_RAW_M ACRO	level	always	known at close t	levels, forward-filled up to 5 business days; use the _ret forms as regressors
DXY_ret,WTI_ret,GOLD_ret,HYG_ret,IEF_ret	05_RAW_M ACRO	log return	always	known at close t	log differences of the above
CreditStress	05_RAW_M ACRO	log return	always	known at close t	$-(HYG_ret - IEF_ret)$; rises with credit stress, same sign as a widening high-yield spread
MacroFilled_t	quality	boolean	always	-	TRUE where at least one macro level was forward-filled on this row

Columns	Group	Units	Availability	Timing	Note
SessionClass	quality	FULL/HALFDAY/DEFECT/MISSING	always	-	intraday session classification driving RV_Valid
RV_Coverage,HasOpen,MaxInteriorGap_blocks	quality	ratio / bool / blocks	where intraday exists	-	inputs to the classification, kept for auditability
RV_Valid	quality	boolean	always	-	THE GATE for every realized measure. TRUE only for FULL sessions with RV>0
IsHalfDay,IsDefect	quality	boolean	always	-	half-days are real short sessions; defects are feed failures
InSample_A,InSample_B,InSample_C	sample	boolean	always	-	sample windows; B is PRIMARY (all six indices, 2013-09-30 onward)
CommonDate_B	sample	boolean	always	-	date present for all six indices in sample B, ignoring RV validity
CrisisLabel,IsCrisis	regime	label / boolean	always	known at close t	10 named crisis windows with fixed calendar dates, identical across all six indices so cross-index comparison is meaningful. See CRISIS_PERIODS.csv for the dates and the basis for each. Sample B contains 6 of the 10; it MISSES the GFC, dot-com, Asian crisis and Euro sovereign crisis
TrailVol63_Ann	regime	annualised %	always	known at close t	trailing 63-day return volatility, annualised. Defined on every day including intraday-defect days, because it uses returns rather than RV
VolRegime	regime	Calm/Normal/Stressed/Crisis	always	EX-POST	quartiles of TrailVol63_Ann using FULL-SAMPLE cut-points. In-sample information: correct for ex-post subsample REPORTING, NEVER as a predictor
VolRegime_ExAnte	regime	Calm/Normal/Stressed/Crisis	from ~750 obs	known at close t	same quartiles from EXPANDING-WINDOW quantiles using only data strictly before each date. No look-ahead - safe as a regressor or conditioning variable
BalancedRV_B	sample	boolean	always	-	date where ALL SIX indices have a VALID realized measure. Use for POOLED cross-index statistics only; per-index work should use RV_Valid and keep its own days